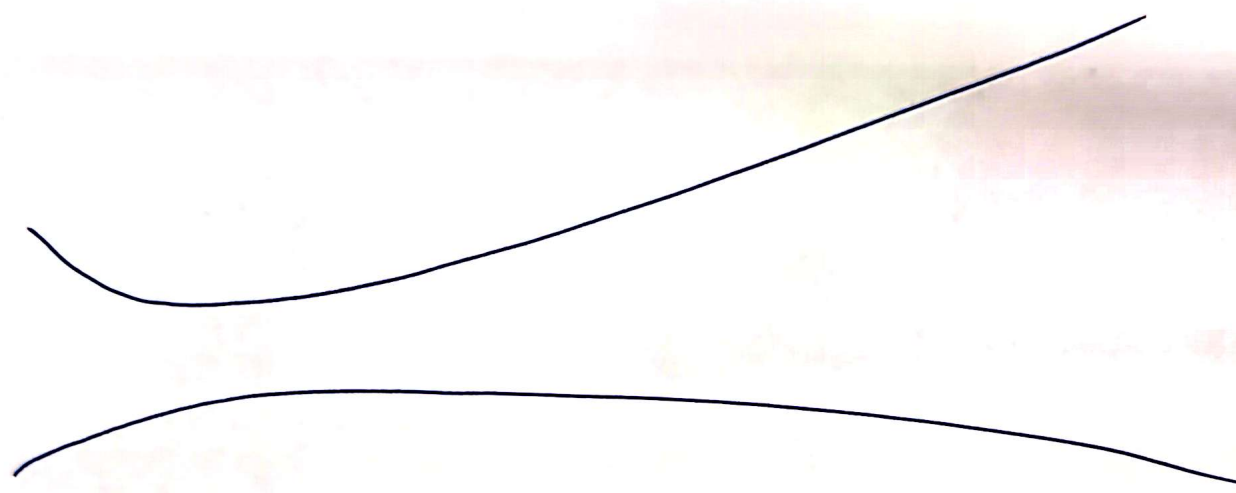




"Method of characteristics"

A method to solve PDE's by
converting them to ODE's

A method to design wave-free nozzles

remember

$M > 1$

region
of silence

region
of
information

Introduction:

- Linearization solutions for the partial differential equation of the velocity potential often lead to inadequate approximations.
- This chapter introduces the method of characteristics (MOC) for solving flow problems described by hyperbolic PDEs, which depends on determination of the direction of the Mach lines or characteristics.

Important properties of characteristics to derive the equations of MOC:

Property 1: A characteristic in a two-dimensional supersonic flow is a curve or line along which physical disturbances are propagated at the local speed of sound relative to the gas.

- Pathlines propagating information of the condition of the flow, such as p , h , and s .
- Acoustic waves travel along Mach lines (or characteristics) propagating information of flow disturbances.

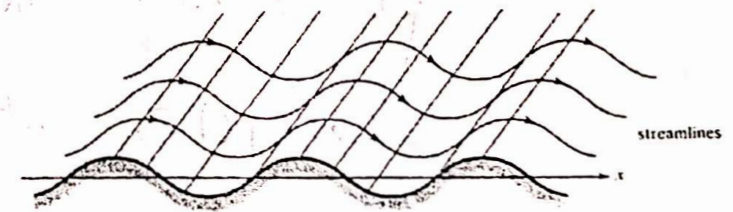
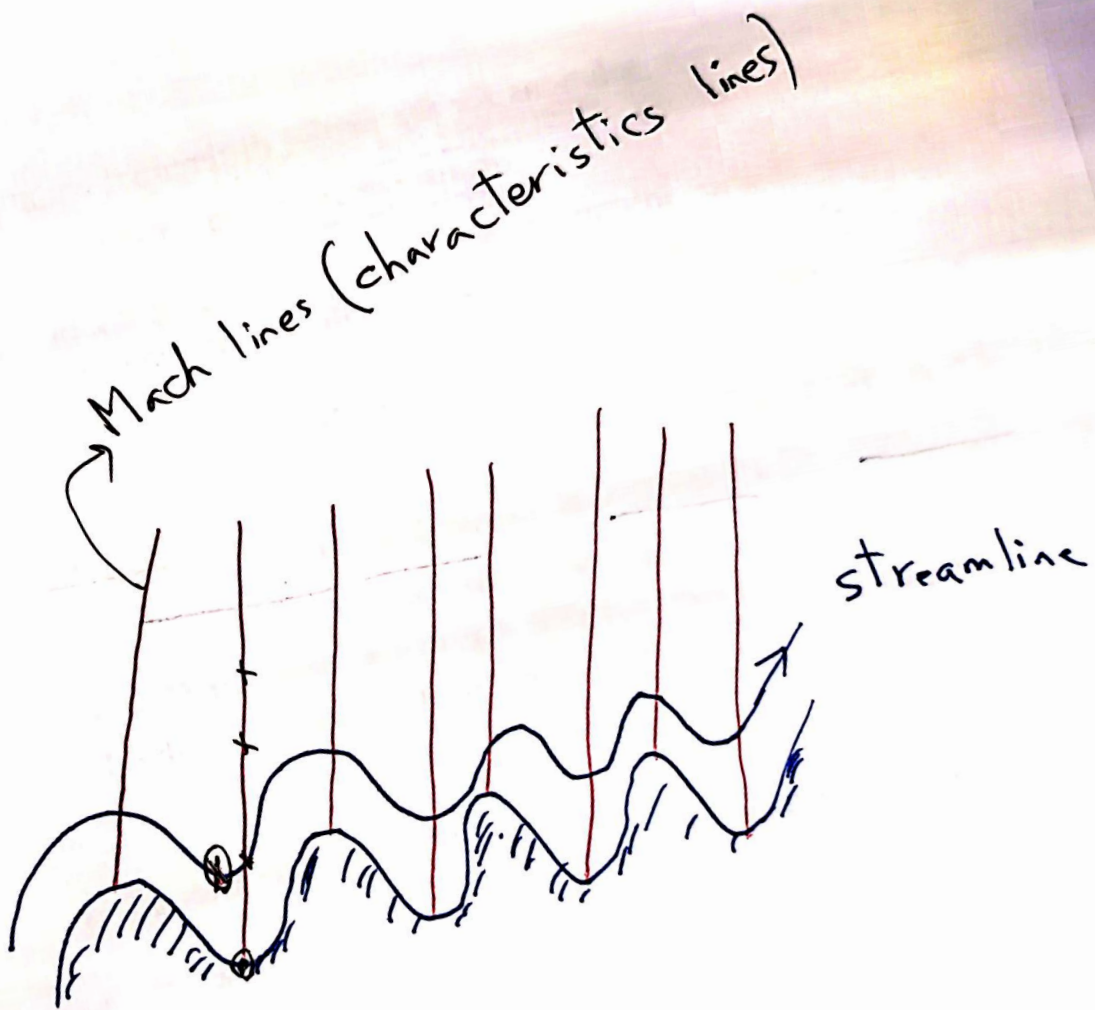


Figure 14.1 Linearized Supersonic Flow over a Wavy Wall

✓



Property 2: A characteristic is a curve across which flow properties are continuous, although they may have discontinuous first derivatives, and along which the derivatives are indeterminate.

❖ A weak disturbance by a small change in wall direction (Prandtl-Meyer):

- The flow behind the Mach line is uniform but has different conditions before and after the wave.
- Ahead of and behind the wave, the velocity derivatives are zero due to uniform flow, but the finite derivative at the wave indicates a discontinuity in the velocity gradient across the Mach line.

□ If there is a discontinuity in the derivative of a variable, but not in the variable itself, the discontinuity is said to be weak. Thus, a Mach line is a weak discontinuity.

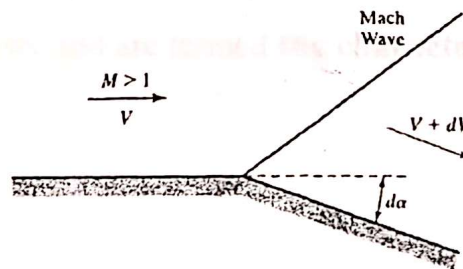
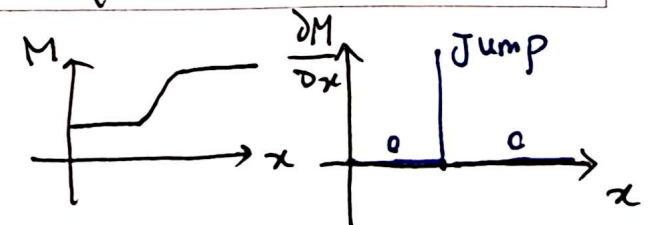


Figure 14.2 Infinitesimal Change of Wall Angle to Create a Mach Wave

$M_1 = \text{cte}$
 $\frac{\partial M_1}{\partial x} = 0$
 $\frac{\partial M}{\partial x} \neq 0$
 $M_2 = \text{cte}$
 $\frac{\partial M_2}{\partial x} = 0$

the first derivative of $M(V)$ is discontinuous



Mach wave
 weak discontinuity

$\left\{ \begin{array}{l} M \text{ is continuous} \\ \frac{\partial M}{\partial x} \text{ " dis "} \end{array} \right.$

given

$$r. I_{u, \dots} \quad (u, \partial u = c) \div a \implies \left\{ \frac{\partial u}{\partial x} + \frac{b}{a} \frac{\partial u}{\partial y} = \frac{c}{a} \right.$$

helping

Property 3:

into an ordina

▪ A slightly

❖ Thre

1- Obtair

2- Determ

3- Solv

$$a \frac{\partial u}{\partial x} + b \frac{\partial u}{\partial y} = c$$

(PDE Eqⁿ) of two unknowns we need one

$$\boxed{dx \frac{\partial u}{\partial x} + dy \frac{\partial u}{\partial y} = du}$$

helping Eqⁿ

[1 less than number of unknowns you need helping Eqⁿ]

$$\begin{bmatrix} a & b \\ dx & dy \end{bmatrix} \begin{bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial y} \end{bmatrix} = \begin{bmatrix} c \\ du \end{bmatrix} \implies$$

$$\left\{ \begin{aligned} \frac{\partial u}{\partial x} &= \frac{\begin{vmatrix} c & b \\ du & dy \end{vmatrix}}{\begin{vmatrix} a & b \\ dx & dy \end{vmatrix}} = \frac{(\frac{c}{a}) \frac{dy}{dx} - (\frac{b}{a}) \frac{du}{dx}}{\frac{dy}{dx} - \frac{b}{a}} \\ \frac{\partial u}{\partial y} &= \frac{\begin{vmatrix} a & c \\ dx & du \end{vmatrix}}{\begin{vmatrix} a & b \\ dx & dy \end{vmatrix}} = \frac{\frac{du}{dx} - \frac{c}{a}}{\frac{dy}{dx} - \frac{b}{a}} \end{aligned} \right.$$

$$\text{if } \frac{dy}{dx} = \frac{b}{a} \longrightarrow \begin{cases} \frac{\partial u}{\partial x} = \frac{\infty}{0} = \infty \\ \frac{\partial u}{\partial y} = \frac{\infty}{0} = \infty \end{cases}$$

charac. lines $(\frac{dy}{dx} = \frac{b}{a})$

$$\text{if } \frac{dy}{dx} = \frac{b}{a} \text{ and if } \frac{du}{dx} = \frac{c}{a} \longrightarrow \begin{cases} \frac{\partial u}{\partial x} = \frac{0}{0} \\ \frac{\partial u}{\partial y} = \frac{0}{0} \end{cases}$$

compatibility Eqⁿ

$$\begin{array}{l}
 \text{given} \rightarrow \left(a \frac{\partial u}{\partial x} + b \frac{\partial u}{\partial y} = c \right) \div a \\
 \text{helping} \rightarrow \left(dx \frac{\partial u}{\partial x} + dy \frac{\partial u}{\partial y} = du \right) \div dx
 \end{array}
 \Rightarrow
 \begin{cases}
 \frac{\partial u}{\partial x} + \frac{b}{a} \frac{\partial u}{\partial y} = \frac{c}{a} \\
 \frac{\partial u}{\partial x} + \frac{dy}{dx} \frac{\partial u}{\partial y} = \frac{du}{dx}
 \end{cases}$$

Property 3: A characteristic is a curve along which the governing partial differential equation(s) may be manipulated into an ordinary differential equation(s). (A characteristic is a curve along which the governing PDE becomes an ODE.)

- A slightly rearranged version of first-order PDE and total derivative :

$$\begin{cases}
 \frac{\partial u}{\partial x} + \frac{b}{a} \frac{\partial u}{\partial y} = \frac{c}{a} \\
 \frac{\partial u}{\partial x} + \frac{dy}{dx} \frac{\partial u}{\partial y} = \frac{du}{dx}
 \end{cases}$$

if $dy/dx = b/a$, as before



$$\boxed{\frac{du}{dx} = \frac{c}{a}}$$

$$\begin{cases}
 \frac{dy}{dx} = \frac{b}{a} \text{ charc. Eq} \\
 \frac{du}{dx} = \frac{c}{a} \text{ comp. Eq}
 \end{cases}$$

❖ Three-step procedure of the MOC:

- 1- Obtain the equation for the slope of the characteristic (dy/dx).
- 2- Determine how the flow variables change along the characteristic (compatibility equation).
- 3- Solve the pair of ordinary differential equations.

$$\frac{dy}{dx} = \frac{b}{a}$$

$$\frac{du}{dx} = \frac{c}{a}$$

$$C_1 \phi_{xx} + 2C_2 \phi_{xy} + C_3 \phi_{yy} = 0 \quad \leftarrow \text{Compressible potential flow Eq}^n$$

$$(1-M_\infty^2) \phi_{xx} + \phi_{yy} = 0 \quad \leftarrow \text{Incomp.} \quad " \quad " \quad " \quad : \nabla^2 \phi = 0$$

Theory of characteristics:

linearized comp.

Three methods available for determining both the slopes of the characteristics and the compatibility equations:

- ❖ The Method of Indeterminate Derivatives (based on Property 2).
- ❖ The Method of Linear Combination (based on Property 3).
- ❖ The Eigen analysis (based on both Properties 2 and 3).

➤ The nonlinear potential equation for two-dimensional isentropic, irrotational steady flow of a perfect gas with constant specific heats:

$$\left[a^2 - \left(\frac{\partial \phi}{\partial x} \right)^2 \right] \frac{\partial^2 \phi}{\partial x^2} - 2 \frac{\partial \phi}{\partial x} \frac{\partial \phi}{\partial y} \frac{\partial^2 \phi}{\partial x \partial y} + \left[a^2 - \left(\frac{\partial \phi}{\partial y} \right)^2 \right] \frac{\partial^2 \phi}{\partial y^2} = 0$$

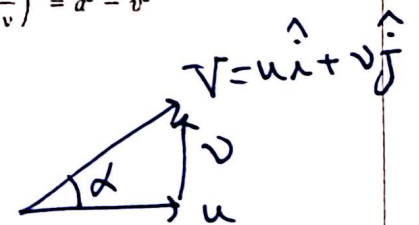
$$C_1 \frac{\partial^2 \phi}{\partial x^2} + 2C_2 \frac{\partial^2 \phi}{\partial x \partial y} + C_3 \frac{\partial^2 \phi}{\partial y^2} = 0$$

$$\begin{cases} C_1 = a^2 - \left(\frac{\partial \phi}{\partial x} \right)^2 = a^2 - u^2 \\ C_2 = -\frac{\partial \phi}{\partial x} \frac{\partial \phi}{\partial y} = -uv \\ C_3 = a^2 - \left(\frac{\partial \phi}{\partial y} \right)^2 = a^2 - v^2 \end{cases}$$

▪ For this isentropic flow, the velocity components, and hence the derivatives, are continuous:

$$d\left(\frac{\partial \phi}{\partial x}\right) = \frac{\partial^2 \phi}{\partial x^2} dx + \frac{\partial^2 \phi}{\partial x \partial y} dy$$

$$d\left(\frac{\partial \phi}{\partial y}\right) = \frac{\partial^2 \phi}{\partial y \partial x} dx + \frac{\partial^2 \phi}{\partial y^2} dy$$



$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

$$dv = \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy$$

$$d\left(\frac{\partial \phi}{\partial x}\right) = \frac{\partial^2 \phi}{\partial x^2} dx + \frac{\partial^2 \phi}{\partial x \partial y} dy$$

$$d\left(\frac{\partial \phi}{\partial y}\right) = \frac{\partial^2 \phi}{\partial x \partial y} dx + \frac{\partial^2 \phi}{\partial y^2} dy$$

$$+ 0 \frac{\partial^2 \phi}{\partial y^2}$$

$$+ 0 \frac{\partial^2 \phi}{\partial x^2}$$

1st helping

2nd helping

Method 1-The Method of Indeterminate Derivatives:

- Along a characteristic, the derivatives of the flow variables are indeterminate (property 2).

Expressions for the characteristics (lines across which there may be a discontinuity in the velocity derivatives or which there is not a unique solution for the velocity derivative):

$$\begin{bmatrix} dx & dy & 0 \\ 0 & dx & dy \\ C_1 & C_2 & C_3 \end{bmatrix} \begin{bmatrix} \phi_{xx} \\ \phi_{xy} \\ \phi_{yy} \end{bmatrix} = \begin{bmatrix} d\frac{\partial\phi}{\partial x} \\ d\frac{\partial\phi}{\partial y} \\ 0 \end{bmatrix}$$

Three linear equations:

$$\begin{cases} dx \frac{\partial^2 \phi}{\partial x^2} + dy \frac{\partial^2 \phi}{\partial x \partial y} + 0 \frac{\partial^2 \phi}{\partial y^2} = d\left(\frac{\partial\phi}{\partial x}\right) \rightarrow \text{helping I} \\ 0 \frac{\partial^2 \phi}{\partial x^2} + dx \frac{\partial^2 \phi}{\partial x \partial y} + dy \frac{\partial^2 \phi}{\partial y^2} = d\left(\frac{\partial\phi}{\partial y}\right) \rightarrow \text{helping II} \\ C_1 \frac{\partial^2 \phi}{\partial x^2} + 2C_2 \frac{\partial^2 \phi}{\partial x \partial y} + C_3 \frac{\partial^2 \phi}{\partial y^2} = 0 \rightarrow \text{given Eq}^n \end{cases}$$

➤ Cramer's rule:

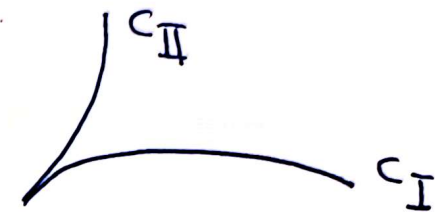
$$|A| \xrightarrow{\text{According to property 2}} 0.0$$

$$|A| = \begin{vmatrix} dx & dy & 0 \\ 0 & dx & dy \\ C_1 & 2C_2 & C_3 \end{vmatrix} = C_3(dx)^2 - 2C_2 dx dy + C_1(dy)^2 \rightarrow C_1\left(\frac{dy}{dx}\right)^2 - 2C_2 \frac{dy}{dx} + C_3 = 0$$

$$\frac{dy}{dx} = \frac{2C_2 \pm \sqrt{4C_2^2 - 4C_1C_3}}{2C_1} = \frac{C_2 \pm \sqrt{C_2^2 - C_1C_3}}{C_1}$$

char. lines

$$\begin{cases} \text{for } C_2^2 - C_1C_3 < 0, dy/dx \text{ will be imaginary} \\ \text{for } C_2^2 - C_1C_3 > 0, dy/dx \text{ will be real.} \end{cases}$$



❖ Replacing C_1, C_2 and C_3 , two characteristics at each point in the flow :

$$\left(\frac{dy}{dx}\right)_{II} = \frac{-\left(\frac{uv}{a^2}\right) - \sqrt{M^2 - 1}}{1 - \left(\frac{u^2}{a^2}\right)} \quad C_{II}$$

$$\left(\frac{dy}{dx}\right)_I = \frac{-\left(\frac{uv}{a^2}\right) + \sqrt{M^2 - 1}}{1 - \left(\frac{u^2}{a^2}\right)} \quad C_I$$

$$M^2 = (u^2 + v^2)/a^2$$

How the velocity components vary with x and y :

➤ Cramer's rule :

$$x_1 = \frac{|A_1|}{|A|}$$



$$\partial u / \partial x = \partial^2 \phi / \partial x^2,$$

$$\frac{\partial^2 \phi}{\partial x^2} = \begin{vmatrix} du & dy & 0 \\ dv & dx & dy \\ 0 & 2C_2 & C_3 \end{vmatrix} = \begin{vmatrix} dx & dy & 0 \\ 0 & dx & dy \\ C_1 & 2C_2 & C_3 \end{vmatrix}$$

Solve for ϕ_{xx} using Cramer's rule

▪ Velocity derivatives must be finite throughout the entire flow field:

$$|A| \xrightarrow{\text{According to property 2}} 0.0$$

$$|A_1| \xrightarrow{\quad} 0.0$$

$$\frac{dv}{du}$$

$$\begin{vmatrix} du & dy & 0 \\ dv & dx & dy \\ 0 & 2C_2 & C_3 \end{vmatrix} = du dx C_3 - du dy 2C_2 - dv dy C_3 = 0$$

$$\frac{dv}{du} = \frac{dx}{dy} - \frac{2C_2}{C_3}$$

$$\frac{dv}{du} = \frac{-C_2 \mp \sqrt{C_2^2 - C_1 C_3}}{C_3}$$

❖ Replacing C1, C2 and C3, the slopes of the characteristics in the hodograph, or velocity plane (u versus v)

$$\left(\frac{dv}{du}\right)_I = \frac{-uv + a\sqrt{u^2 + v^2 - a^2}}{v^2 - a^2}$$

$$\left(\frac{dv}{du}\right)_{II} = \frac{-uv - a\sqrt{u^2 + v^2 - a^2}}{v^2 - a^2}$$

❖ Slope of the characteristic lines in polar coordinates (V, α) (α is the flow inclination):

$$\begin{aligned} \frac{dy}{dx} &= \frac{-uv \pm a\sqrt{u^2 + v^2 - a^2}}{a^2 - u^2} = \frac{\left(\frac{v}{u}\right)\left(\frac{M^2}{M^2 - 1}\right) \mp \left(\frac{a^2}{u^2}\sqrt{M^2 - 1}\right)\left(\frac{M^2}{M^2 - 1}\right)}{\left(\frac{M^2}{M^2 - 1}\right) - \left(\frac{a^2}{u^2}\right)\left(\frac{M^2}{M^2 - 1}\right)} \\ &\quad \times \frac{M^2/[u^2(M^2 - 1)]}{M^2/[u^2(M^2 - 1)]} \end{aligned} \quad (14.40)$$

$$\left[\begin{array}{l} M^2/(M^2 - 1) = 1 + 1/(M^2 - 1) \\ (u^2 + v^2)/u^2 = 1 + v^2/u^2 \\ M^2 = (u^2 + v^2)/a^2 \end{array} \right] \Rightarrow \frac{dy}{dx} = \frac{\left(\frac{v}{u} \mp \frac{1}{\sqrt{M^2 - 1}}\right)}{\left(1 \pm \frac{v}{u} \frac{1}{\sqrt{M^2 - 1}}\right)}$$

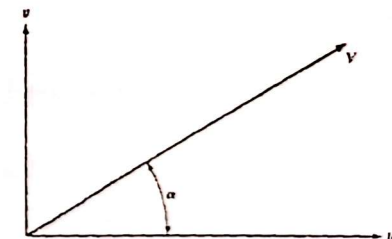
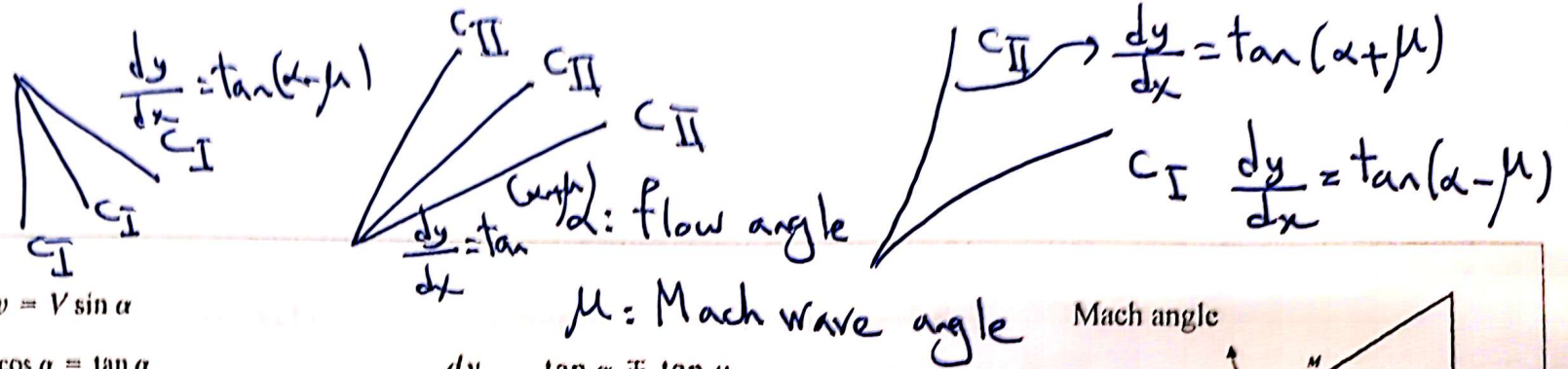


Figure 14.3 Polar Coordinates V and α in the Hodograph Plane



$$\begin{cases} u = V \cos \alpha, v = V \sin \alpha \\ v/u = V \sin \alpha / V \cos \alpha = \tan \alpha \\ 1/\sqrt{M^2 - 1} = \tan \mu \\ \sin \mu = 1/M = a/V \end{cases}$$



$$\frac{dy}{dx} = \frac{\tan \alpha \mp \tan \mu}{1 \pm \tan \alpha \tan \mu} = \tan(\alpha \mp \mu)$$

$$\left(\frac{dy}{dx} \right)_I = \tan(\alpha - \mu) \quad \left(\frac{dy}{dx} \right)_{II} = \tan(\alpha + \mu)$$

Mach angle

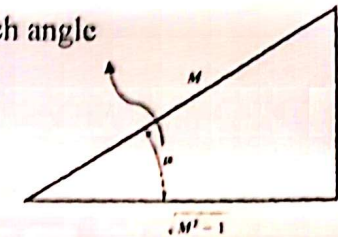


Figure 14.4 Mach Triangle

- The characteristic lines are inclined at the Mach angle to the flow direction.
- Characteristics of Type I (right running) making a negative acute angle with the flow direction
- Characteristics of Type II (left running) making a positive acute angle with the flow direction.

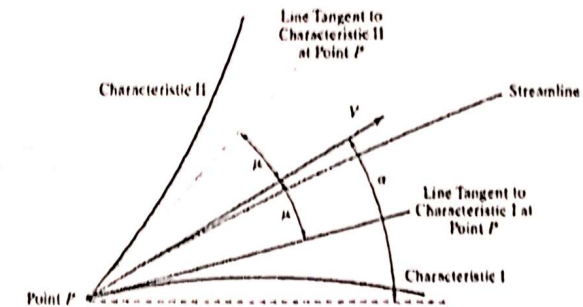
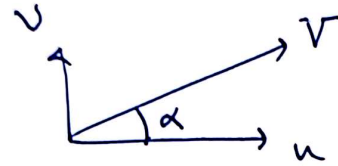


Figure 14.5 Geometry of Characteristics at a Point



$$u = V \cos \alpha$$

$$v = V \sin \alpha$$

$$\frac{v}{u} = \tan \alpha$$

$$\frac{1}{\sqrt{M^2 - 1}} = \tan \mu$$

Mach angle

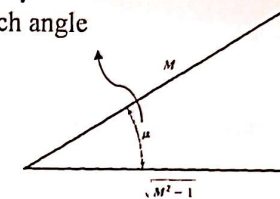


Figure 14.4 Mach Triangle

$$\begin{cases} u = V \cos \alpha, v = V \sin \alpha \\ v/u = V \sin \alpha / V \cos \alpha = \tan \alpha \\ 1/\sqrt{M^2 - 1} = \tan \mu \\ \sin \mu = 1/M = a/V \end{cases}$$



$$\frac{dy}{dx} = \frac{\tan \alpha \mp \tan \mu}{1 \pm \tan \alpha \tan \mu} = \tan(\alpha \mp \mu)$$

$$\left(\frac{dy}{dx} \right)_I = \tan(\alpha - \mu) \quad \left(\frac{dy}{dx} \right)_{II} = \tan(\alpha + \mu)$$

- The characteristic lines are inclined at the Mach angle to the flow direction.
- Characteristics of Type I (right running) making a negative acute angle with the flow direction
- Characteristics of Type II (left running) making a positive acute angle with the flow direction.

α : flow angle
 μ : Mach angle

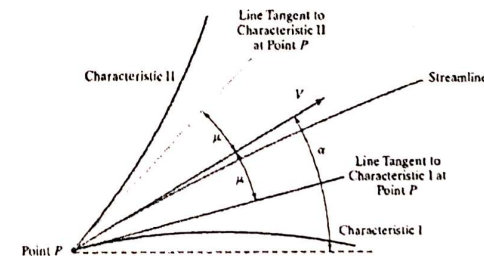
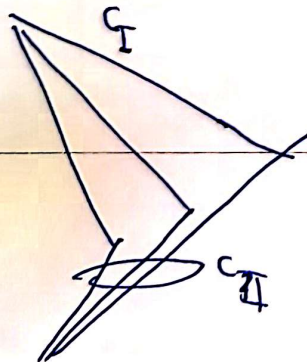


Figure 14.5 Geometry of Characteristics at a Point

- Mach number as a function of α along characteristics(similar integral in dealing with Prandtl-mayer flow)

$$\Delta \nu = \int \sqrt{M^2 - 1} \frac{dV}{V} \quad d\nu = \sqrt{M^2 - 1} \frac{dV}{V}$$

$$\boxed{\frac{1}{V} \frac{dV}{d\alpha} = + \frac{1}{\sqrt{M^2 - 1}}}$$

$$\int_a^b d\alpha = - \int_a^b \sqrt{M^2 - 1} \frac{dV}{V} \quad \text{along a characteristic of Type I}$$

$$\boxed{\frac{1}{V} \frac{dV}{d\alpha} = - \frac{1}{\sqrt{M^2 - 1}}}$$

$$\int_a^b d\alpha = + \int_a^b \sqrt{M^2 - 1} \frac{dV}{V} \quad \text{along a characteristic of Type II}$$

- The A and b are points on the characteristics:

$$\alpha_b - \alpha_a = -(\nu_b - \nu_a)$$

- Let a be a reference state, so that :

$$\alpha + \nu = \alpha_{\text{ref}} + \nu_{\text{ref}}$$

Reimann invariants

$$\nu - \alpha = \text{constant} = C_{II}$$

$$\nu + \alpha = \text{constant} = C_I$$

P.M function

Handwritten diagram showing characteristics and slopes:

For Type II characteristic (labeled C_{II}):

$$\frac{dy}{dx} = \tan(\alpha + \mu)$$

For Type I characteristic (labeled C_I):

$$\frac{dy}{dx} = \tan(\alpha - \mu)$$

Along a Type I characteristic, $\nu + \alpha = \text{cte}$.